NBA non-contact injury survival — Update 2: baseline models

Survival dataset: 229,064 player-game landmarks, 1,353 players; follow-up up to 30 days; 42 features

Training 2016-17..2021-22: 160,783 landmarks (5,425 ending in a target injury)

Validation 2022-23: 27,740 landmarks (1,472 ending in a target injury)

Test season 2023-24 not used (sealed until final model selection).

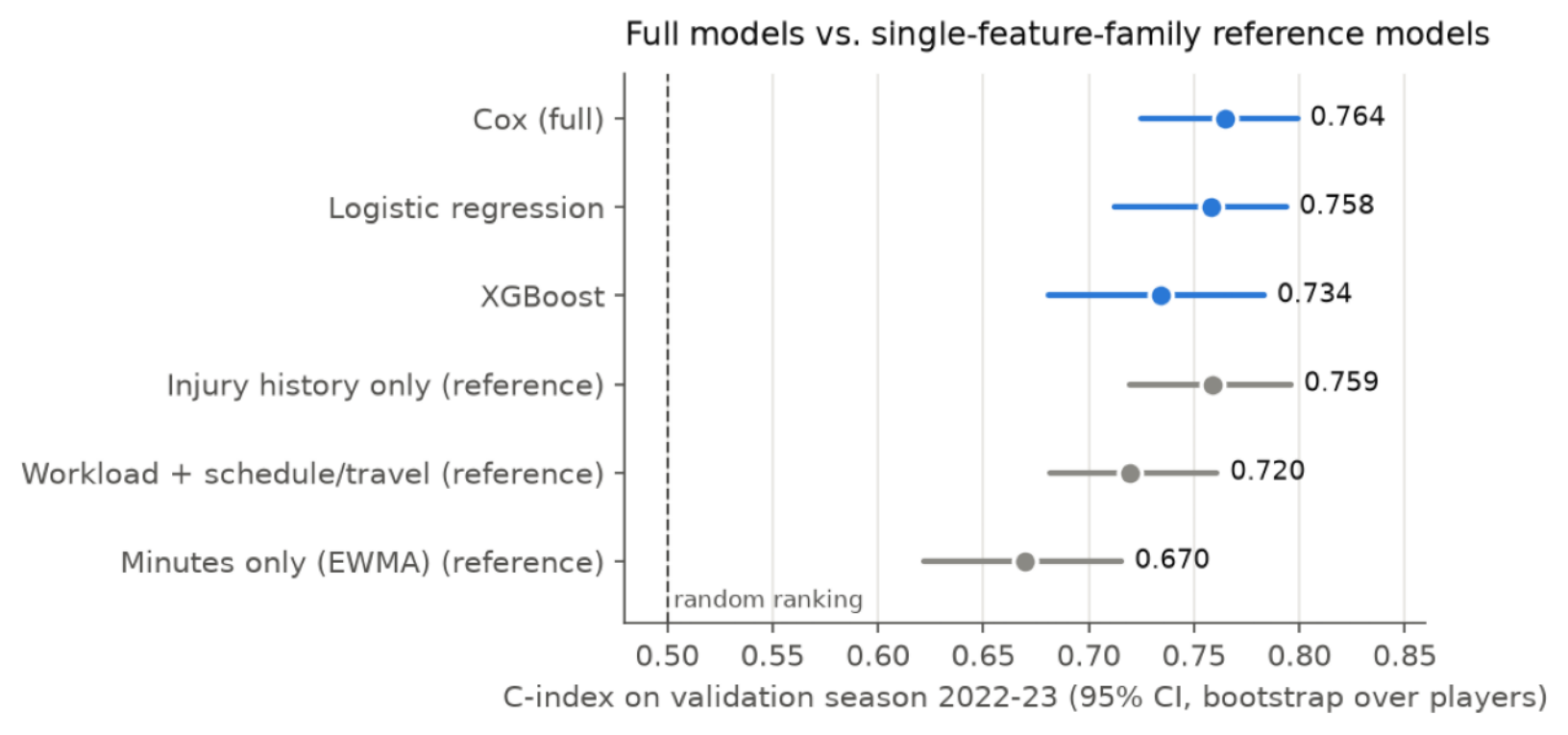
C-index on validation season (95% CI)

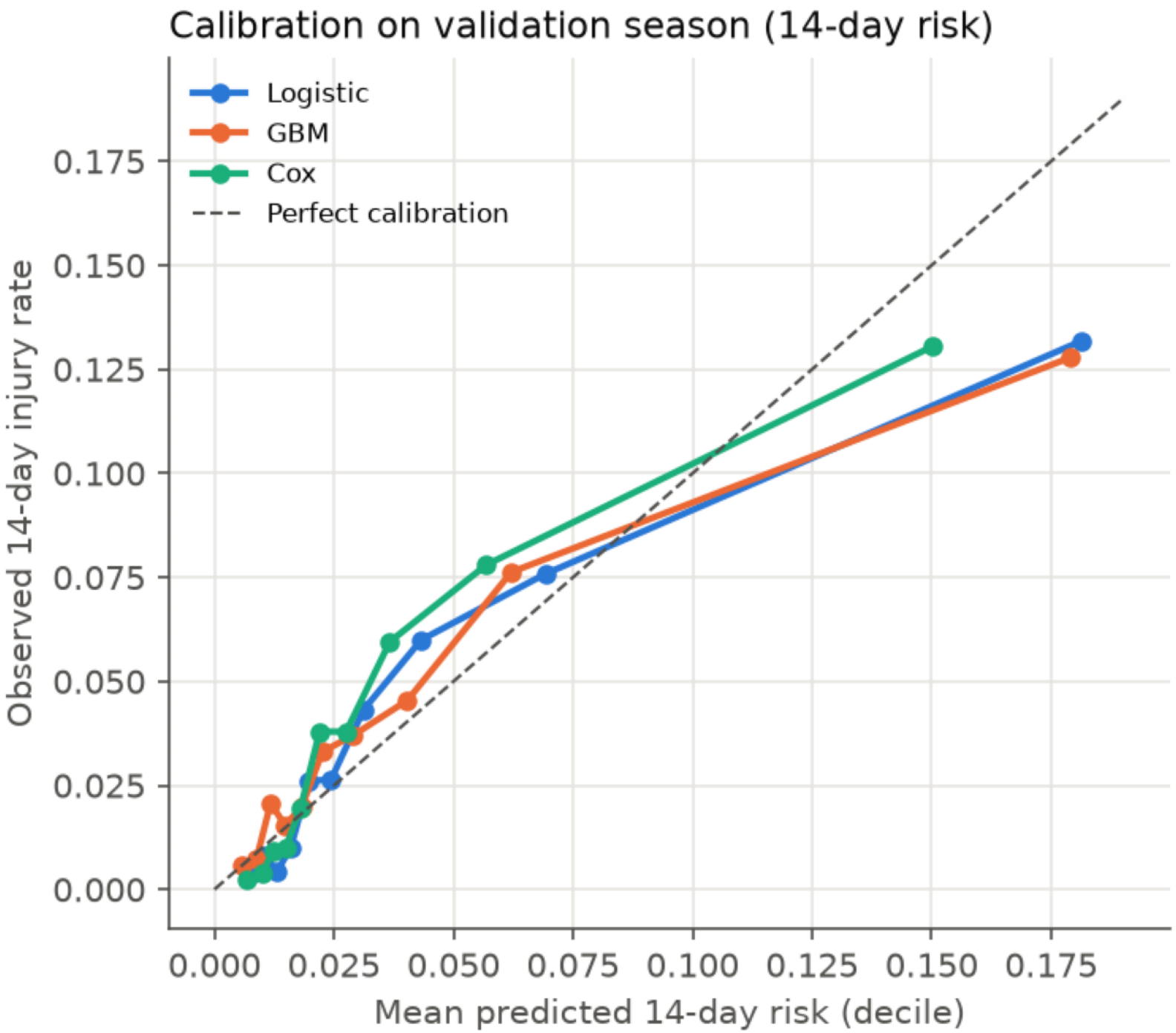
Cox (full):	0.764 (0.725–0.799)
Logistic regression:	0.758 (0.712–0.794)
XGBoost:	0.734 (0.681–0.783)

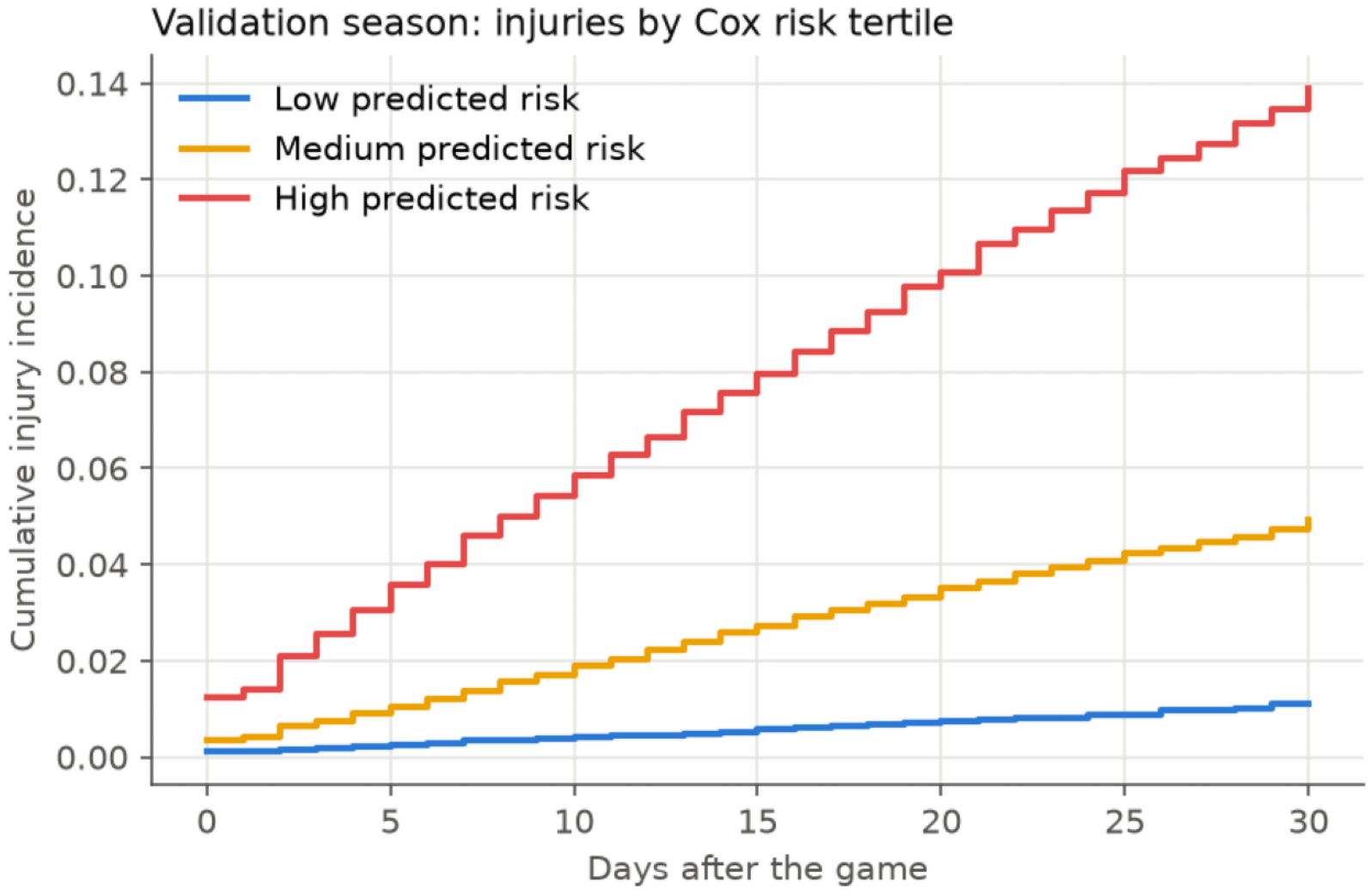
Injury history alone: 0.759; workload + schedule/travel alone: 0.720; minutes alone: 0.670

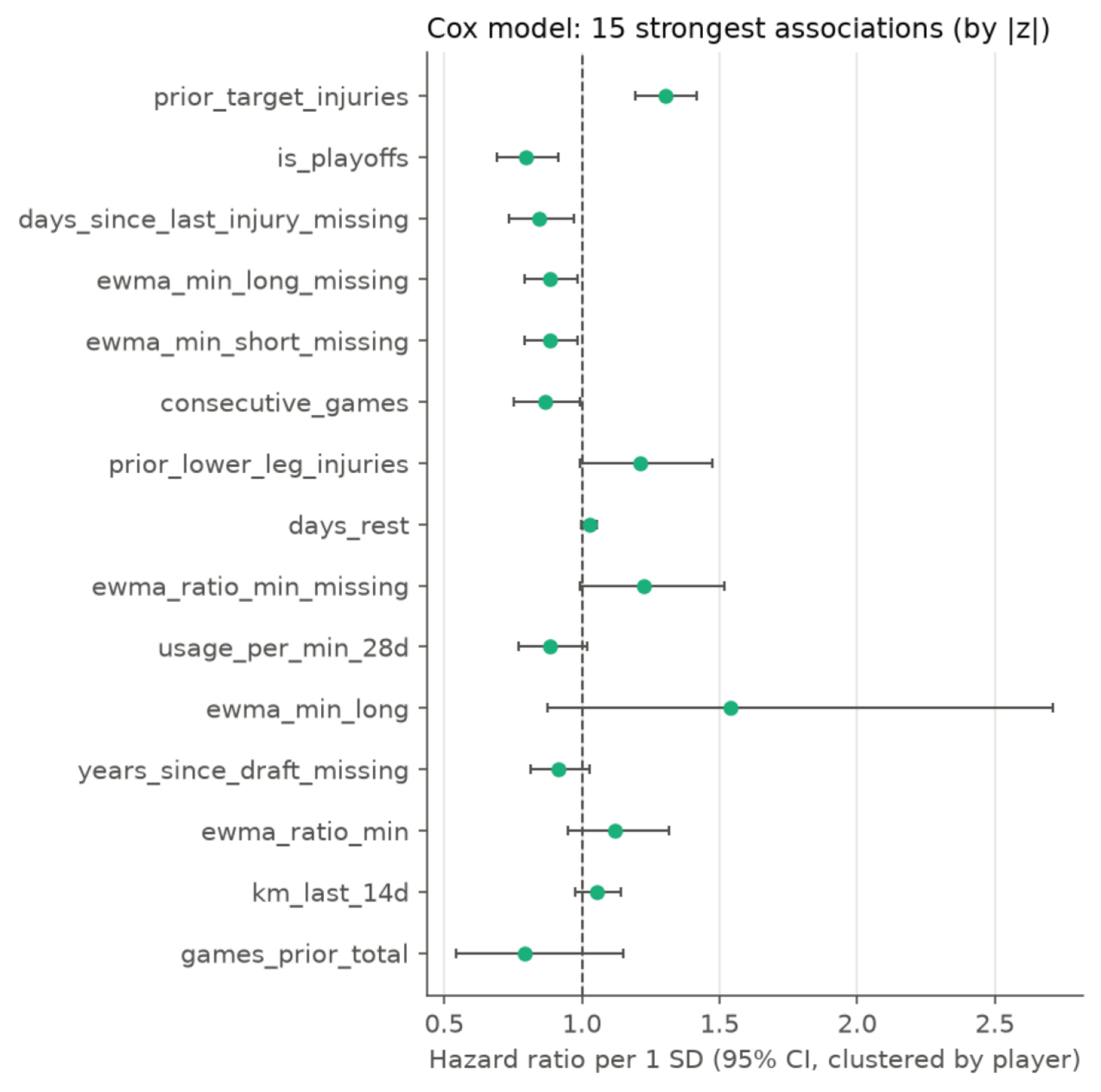
-> The models mainly identify which players get hurt, not when.

Sensitivity analyses (train from 2018-19; exclude 2019-21) change C-index by at most 0.006.


Full models vs. single-feature-family reference models
Cox (full)
Logistic regression
XGBoost
Injury history only (reference)
Workload + schedule/travel (reference)
Minutes only (EWMA) (reference)
0.764
0.758
0.734
0.759
0.720
0.670
random ranking
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
C-index on validation season 2022-23 (95% CI, bootstrap over players)


Calibration on validation season (14-day risk)
Logistic
GBM
Cox
Perfect calibration
Observed 14-day injury rate
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Mean predicted 14-day risk (decile)


Validation season: injuries by Cox risk tertile
Low predicted risk
Medium predicted risk
High predicted risk
Cumulative injury incidence
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
5
10
15
20
25
30
Days after the game


Cox model: 15 strongest associations (by |z|)
prior_target_injuries
is_playoffs
days_since_last_injury_missing
ewma_min_long_missing
ewma_min_short_missing
consecutive_games
prior_lower_leg_injuries
days_rest
ewma_ratio_min_missing
usage_per_min_28d
ewma_min_long
years_since_draft_missing
ewma_ratio_min
km_last_14d
games_prior_total
0.5
1.0
1.5
2.0
2.5
Hazard ratio per 1 SD (95% CI, clustered by player)